Lab Handout 1-A
Radio Frequency (RF) Breakout Assembly and First Power-On

Capstone — Module 1: RF Hardware Foundations

1 Objective

Transform a bag of loose parts (Adafruit RFM95W breakout, 0.1 " header strip, 915 MHz spring antenna) into a breadboard-ready, power-verified RF module. By the end of this handout you will have a physically assembled breakout that draws the correct idle current and is ready for Serial Peripheral Interface (SPI) bring-up in Lab Handout 1-C.

2 Prerequisites

- No prior lab handouts required.
- Basic through-hole soldering experience.

3 Materials

Item	Qty	Notes
Adafruit RFM95W 900 MHz LoRa breakout	1	PID 3072 or equivalent
0.1 " male header strip (snappable)	1	Ships with breakout
915 MHz spring (helical) antenna	1	Bare coil, single solder lead
Soldering iron (temperature-controlled)	1	320–350 °C tip
Rosin-core solder (0.6–0.8 mm, 60/40 or lead-free)	—	
Flux pen or paste (optional)	—	Recommended for clean joints
Flush cutters	1	For trimming header and antenna leads
Breadboard	1	Used as a soldering jig
Bench DMM (voltage + current ranges)	1	For power-on verification
Bench power supply *or* Nucleo 3.3 V rail	1	3.3 V, current-limited to 100 mA

4 Procedure

4.1 Step 1 — Snap and install the header strip

1. Count the pins on the long edge of the breakout (14 for the RFM9x breakout). Snap off a matching length of header strip.
2. Insert the long pins of the header strip down into a breadboard. The short pins point up. This is your soldering jig — it keeps the header perpendicular to the board while you solder.
3. Drop the breakout PCB onto the upward-facing short pins, component-side up. The pins should poke through the row of plated holes along the edge.

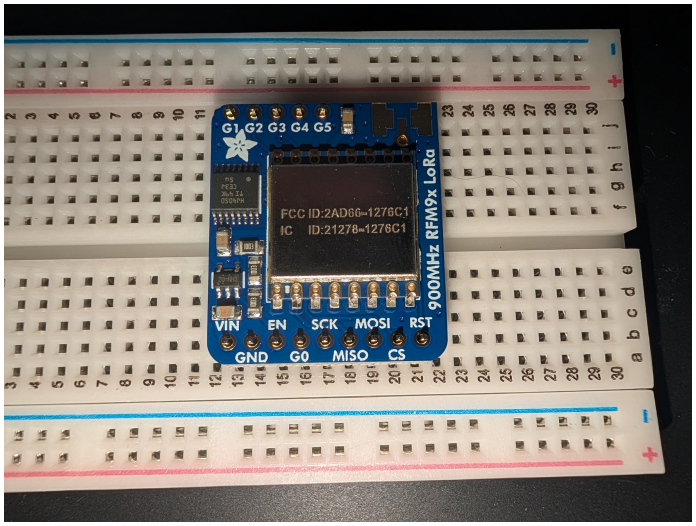

Figure 1: RFM95W breakout dry-fit onto the snapped header strip, using the breadboard as a right-angle soldering jig.

4.2 Step 2 — Solder the header pins

1. Heat the iron fully. Apply a small amount of fresh solder to the tip, then wipe on a brass sponge (tinning).
2. Solder each header pin one at a time. Touch the iron tip to the junction between the pin and the pad for ∼1 second, feed in a small amount of solder, wait ∼1 second for it to flow, then remove the solder, then remove the iron.
3. A good joint is shiny, concave, and fully wets both the pin and the pad. A bad joint is dull, ball-shaped, or only touches one side.
4. Inspect between every 3–4 pins to catch solder bridges before they get buried.

Common Pitfalls

- **Cold joints:** pulling the iron too soon. Count one-thousand-one, one-thousand-two before lifting.
- **Bridges:** two adjacent pins connected by a blob. Remove with desoldering wick or flux + reheat.
- **Lifted pads:** from reheating the same joint repeatedly. If you need to redo a joint, wait for it to cool to room temperature first.

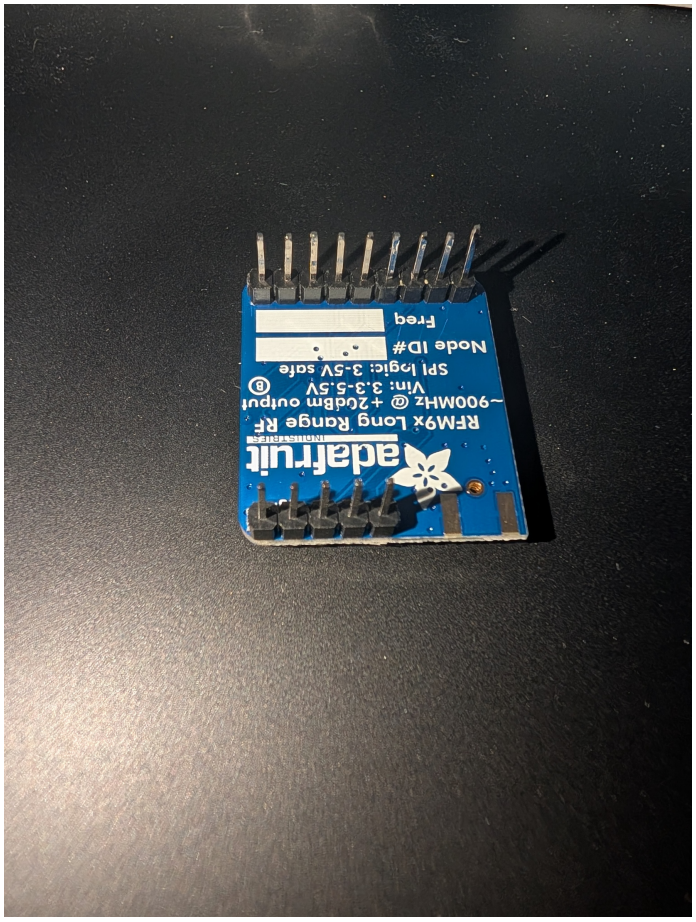

Figure 2: Macro view of finished header joints — shiny, concave fillets that fully wet pin and pad. Use this as a pass/fail reference.

4.3 Step 3 — Install the spring antenna

1. Locate the `ANT` through-hole on the breakout (labeled on the silkscreen, near one corner).
2. Insert the single lead of the spring antenna into the `ANT` hole from the top side. The coil should stand vertical.
3. Hold the antenna perpendicular to the board. Solder the lead on the bottom side. Use slightly more solder than a header pin — this joint is the only mechanical anchor for the coil.
4. Trim the excess lead on the bottom side with flush cutters, leaving no more than 1 mm protruding.

Why the spring, not the SMA?

This course uses the direct-soldered spring antenna instead of the edge-launch SMA option. The spring is shorter, lighter, and eliminates a connector interface that is a common source of intermittent faults during benchtop work. The SMA option is reserved for Module 5 range testing, where the antenna may need to be physically separated from the board.

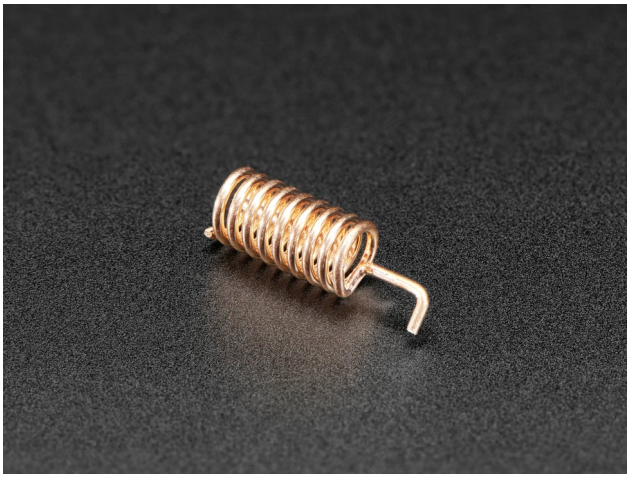

Figure 3: Spring antenna soldered vertically into the `ANT` pad. The coil is the sole mechanical anchor, so use slightly more solder than a header pin.

4.4 Step 4 — Visual inspection

1. Hold the board under a bright light and a magnifier (phone camera zoom works).
2. Verify: no solder bridges between pins; no cold (dull) joints; no flux residue extending beyond the pads; the antenna coil is mechanically stable and stands upright.
3. Clean flux residue with isopropyl alcohol and a soft brush if used.

4.5 Step 5 — First power-on

1. Insert the assembled breakout into a breadboard. Leave at least one empty column on each side for jumper routing.
2. Wire `GND` to the supply ground and `VIN` to a *current-limited* 3.3 V source (bench supply set to 100 mA compliance, or the Nucleo's 3.3 V rail through a DMM in current mode).
3. Apply power. The module should draw approximately **1 mA** in idle (post-reset, SX1276 in `SLEEP` mode). Anything >20 mA at this stage indicates a short; remove power immediately and re-inspect.
4. With a DMM in voltage mode, confirm that the `3V3` test point on the breakout reads 3.30 ± 0.15 V. This validates the on-board regulator.

[Figure placeholder]

`LH1A_idle_current.jpg`

Bench DMM in series with the 3.3 V rail, reading ≈1 mA idle current. Anything above 20 mA at this stage indicates a short.

Figure 4: Bench DMM in series with the 3.3 V rail, reading ≈1 mA idle current. Anything above 20 mA at this stage indicates a short.

5 Verification Checklist

- ☐ All 14 header pins soldered with clean, shiny fillets.
- ☐ No solder bridges between any two adjacent pins.
- ☐ Spring antenna mechanically stable, vertical, and electrically continuous to the `ANT` pad (DMM continuity test from coil tip to `ANT` pad).
- ☐ Idle current draw at 3.3 V is within 0.5–5 mA.
- ☐ On-board regulator output measures 3.15–3.45 V.

Lab Handout 1-A Deliverable

Submit the following before beginning Lab Handout 1-C:

1. A clear photograph of the assembled breakout (top and bottom).
2. Measured idle current draw at 3.3 V, in mA.
3. Measured `3V3` regulator output, in V.
4. One-sentence pass/fail statement for each verification checklist item.

Glossary

RF	Radio Frequency	1
SPI	Serial Peripheral Interface	1